

Stochastic Local Search Algorithms for the Linear Ordering Problem

INFO-H-413 — Heuristic Optimization — Implementation Exercise 2

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Abstract

We design and evaluate two stochastic local search algorithms for the Linear Ordering Problem: Simulated Annealing (SA) and a MAX-MIN Ant System (MMAS). Both build on the VND local search from the first implementation exercise. SA explores the search space through random insert moves with a temperature-controlled acceptance criterion. MMAS constructs solutions using pheromone trails and heuristic information, then refines each one with VND. Both algorithms are evaluated on all instances of size 150 and compared using Wilcoxon signed-rank tests and run-time distributions.

1. Introduction

In the first implementation exercise, we studied iterative improvement algorithms for the Linear Ordering Problem (LOP). We found that Insert with first-improvement and CW initialisation gives the best single-neighbourhood results, and that VND-TIE slightly improves on it. But all these methods stop at the first local optimum they reach. There is no mechanism to escape and keep searching.

This exercise addresses that limitation. We implement two stochastic local search (SLS) algorithms that can escape local optima and continue exploring the search space over a longer time budget. The two methods come from different classes: Simulated Annealing is a simple SLS method, and MAX-MIN Ant System is a population-based method. Both use VND from the first exercise as a building block.

The question we are trying to answer is whether these more sophisticated methods can find significantly better solutions than plain local search, and how they compare to each other.

2. Algorithms

2.1. Simulated Annealing

Simulated Annealing (SA) is inspired by the annealing process in metallurgy [1]. The idea is to allow the search to accept worsening moves with some probability, so that it can escape local optima. That probability decreases over time: early on the algorithm explores freely, later it becomes more selective and converges.

2.1.1. Initial solution

We start from a CW solution (Chenery–Watanabe heuristic, as described in the first exercise). CW ranks elements by their row-sum score and gives a reasonable starting point without any local search overhead. We do not apply VND before SA starts, so the full time budget is available for the SA search itself.

2.1.2. Neighbourhood and move generation

We use the **insert neighbourhood**: pick a random element at position i and reinsert it at a random position $j \neq i$. This was the strongest neighbourhood in the first exercise and it gives SA enough freedom to rearrange the permutation significantly in a single step. At each iteration, one random insert move is sampled. The gain Δ is computed using the $O(n)$ delta evaluation from the first exercise.

2.1.3. Acceptance criterion

If the move improves the solution ($\Delta > 0$), it is always accepted. If it worsens it ($\Delta < 0$), it is accepted with probability:

$$P(\text{accept}) = \exp\left(\frac{\Delta}{T}\right)$$

where T is the current temperature. When T is high, the exponential is close to 1 and almost everything is accepted. When T is low, only small degradations have a chance.

2.1.4. Temperature schedule

The initial temperature T_0 is calibrated automatically. Before starting the main loop, we sample 500 random insert moves and compute the average absolute value of the worsening deltas. We set T_0 so that roughly 50% of those moves would be accepted:

$$T_0 = \frac{|\overline{\Delta^-}|}{\ln(2)}$$

This avoids having to guess a problem-specific temperature. The cooling follows a geometric schedule: after each temperature level, we multiply T by a cooling rate $\alpha = 0.99$. Each temperature level consists of $L = n \times 10$ random moves, where n is the instance size. This gives a slow enough cooling that the algorithm has time to explore at each temperature.

2.1.5. Final refinement

When the time limit is reached, we take the best solution found during the entire SA run and apply VND-TEI to it. This ensures that the final solution is at least a local optimum with respect to all three neighbourhoods.

2.1.6. Summary

- **Initialisation:** CW heuristic
- **Neighbourhood:** Insert (random move at each step)
- **Acceptance:** Metropolis criterion with Boltzmann probability
- **Cooling:** Geometric ($\alpha = 0.99$), $L = 10n$ moves per level
- **Calibration:** T_0 set from average worsening delta (50% acceptance)
- **Post-processing:** VND on the best solution found

2.2. MAX–MIN Ant System

Ant Colony Optimisation (ACO) is a population-based metaheuristic inspired by the foraging behaviour of ants [2]. A colony of artificial ants constructs solutions probabilistically, guided by pheromone trails that encode information from previous good solutions. We use the MAX–MIN Ant System (MMAS) variant [3], which bounds the pheromone values to avoid premature convergence.

2.2.1. Pheromone representation

We maintain a pheromone matrix $\tau \in \mathbb{R}^{n \times n}$, where τ_{ij} represents the learned desirability of placing element i before element j in the permutation. All entries are initialised to $\tau_{\max}$ at the start, which encourages exploration in the first iterations.

2.2.2. Construction phase

Each ant builds a complete permutation by placing elements one at a time. At each step, the ant has a set R of remaining (unplaced) elements. For each candidate $j \in R$, we compute:

- A **pheromone score**: $\varphi(j) = \sum_{k \in R, k \neq j} \tau_{jk}$, which measures how much past experience favours placing j before the other remaining elements.
- A **heuristic score**: $\eta(j) = \sum_{k \in R, k \neq j} c_{jk}$, which is the direct benefit of placing j before the remaining elements according to the cost matrix.

The probability of choosing element j is then:

$$p(j) = \frac{[\varphi(j)]^\alpha \cdot [\eta(j)]^\beta}{\sum_{l \in R} [\varphi(l)]^\alpha \cdot [\eta(l)]^\beta}$$

where α and β control the relative importance of pheromone versus heuristic information. We use $\alpha = 1$ and $\beta = 3$, giving more weight to the problem-specific heuristic while still letting pheromone guide the search toward previously successful regions.

2.2.3. Local search

After construction, each ant’s solution is improved using **VND-TEI** (Transpose $\rightarrow$ Exchange $\rightarrow$ Insert, first-improvement). This is the most important step: the constructed solution is typically far from a local optimum, and VND brings it to a much better quality. Without this step, ACO would not be competitive with SA.

2.2.4. Pheromone update

After all ants have constructed and improved their solutions, the pheromone matrix is updated in two phases:

Evaporation. All pheromone values are reduced by a factor $(1 - \rho)$:

$$\tau_{ij} \leftarrow (1 - \rho) \cdot \tau_{ij}$$

This gradually forgets old information and prevents the accumulation of pheromone on suboptimal edges. We use $\rho = 0.2$.

Deposit. The best-so-far solution reinforces the pheromone on its edges. For every pair of positions (p, q) with $p < q$ in the best permutation:

$$\tau_{s_p^*, s_q^*} \leftarrow \tau_{s_p^*, s_q^*} + 1$$

Using the best-so-far solution (rather than the iteration-best) provides a stronger learning signal and pushes the colony toward the best known region of the search space.

Bounds. After each update, all pheromone values are clamped to $[\tau_{\min}, \tau_{\max}]$. This is the core mechanism of MMAS: it prevents any trail from becoming so dominant that the construction becomes deterministic, and it prevents unused trails from vanishing completely.

2.2.5. Parameters

Parameter	Value	Role
m (ants)	10	Number of solutions constructed per iteration
α	1.0	Pheromone weight in construction probability
β	3.0	Heuristic weight in construction probability
ρ	0.2	Evaporation rate
$\tau_{\max}$	10.0	Upper bound on pheromone values
$\tau_{\min}$	0.1	Lower bound on pheromone values

Table 1: ACO parameter settings.

The number of ants $m = 10$ is a compromise between diversity (more ants explore more of the search space per iteration) and cost (each ant requires a full VND run). With $n = 150$, each VND call takes a non-negligible amount of time, so using too many ants would leave too few iterations for pheromone learning.

The choice $\beta = 3$ gives a strong bias toward the heuristic information. This makes sense because the CW-style heuristic (sum of row weights) is already a good predictor of element quality. The pheromone then fine-tunes the ordering based on experience.

2.2.6. Summary

- **Construction:** Probabilistic, guided by pheromone (τ) and heuristic (η)
- **Local search:** VND-TEI applied to every constructed solution
- **Pheromone update:** Evaporation ($\rho = 0.2$) + deposit on best-so-far
- **Bounds:** MMAS with $\tau \in [0.1, 10.0]$
- **Colony size:** 10 ants per iteration

3. Experimental Setup

3.1. Instances and termination criterion

Both algorithms were run once on each instance of size 150. The termination criterion is defined as:

$$t_{\max} = \overline{t_{\text{VND}}} \times 500$$

where $\overline{t_{\text{VND}}}$ is the average computation time of a full VND run on instances of the same size, re-measured in the experimental environment used for this exercise (**Average VND time : 0.733433s**). This yields a time budget of **Time limit (x500): 366.72s** per run, long enough for the SLS algorithms to perform a meaningful search.

3.2. Implementation

The solver is written in C, compiled with GCC and `-O3`. Because the full experimental campaign (39 instances $\times$ 2 algorithms + 100 RTD runs) would have taken several days on a laptop, all measurements were carried out on a dedicated DigitalOcean Droplet (2 vCPU, Ubuntu 24.04) using a `tmux` session for persistence. Both SA and ACO reuse the VND, delta evaluation functions and instance reader from the first exercise. The random number generator is the same pseudo-random generator from Numerical Recipes used throughout the project. Seeds can be set via the `--seed` command-line flag to allow reproducible runs.

3.3. Evaluation metric

We use the same **Relative Percentage Deviation** (RPD) as in the first exercise:

$$\text{RPD} = \frac{\text{BestKnown} - \text{FinalCost}}{\text{BestKnown}} \times 100$$

Lower is better. An RPD of 0 means we matched the best-known solution.

4. Results and Analysis

4.1. Exercise 2.1 — SLS Results on Size 150 Instances

Table 2 reports the main results for both SLS algorithms across all 39 instances of size 150, together with the best VND configuration (VND-TIE) from Exercise 1 used as a baseline.

Algorithm	Avg RPD	SD	Min RPD	Max RPD	Avg time (s)
Simulated Annealing	0.435	0.160	0.152	0.817	366.72
ACO (MMAS)	0.910	0.193	0.556	1.405	370.62
VND-TIE (Ex. 1)	1.690	0.485	0.828	2.747	0.41

Table 2: Average RPD (%) over all 39 size-150 instances for each SLS algorithm and the best VND baseline from Exercise 1.

Figure 1 and Figure 2 summarise the same information graphically.

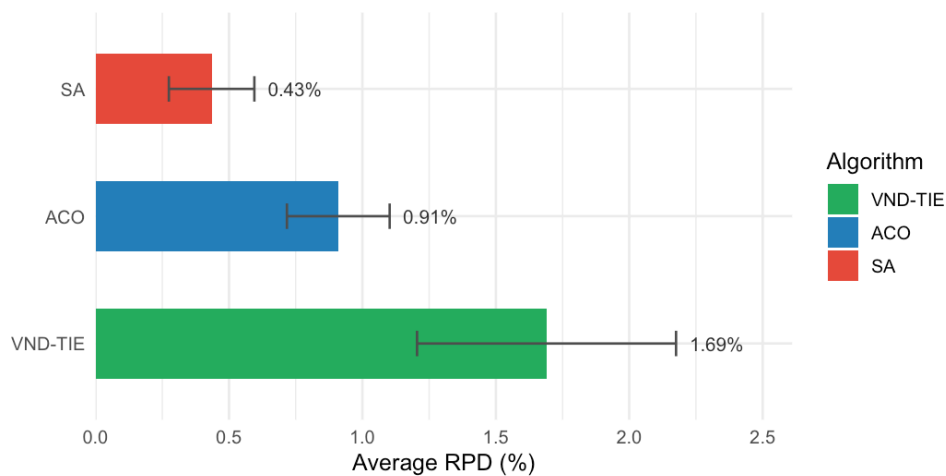


Figure 1: Average RPD (%) for the two SLS algorithms and the VND-TIE baseline from Exercise 1. Error bars represent one standard deviation.

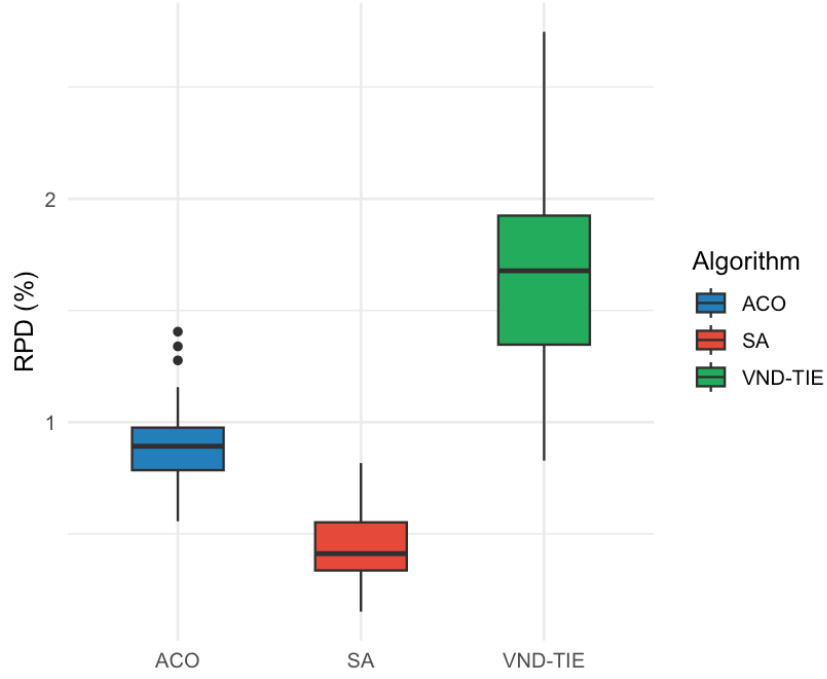


Figure 2: Distribution of RPD values across the 39 size-150 instances for each algorithm.

4.1.1. Correlation Plot

Figure 3 plots, for each of the 39 instances, the SA RPD on the x -axis against the ACO RPD on the y -axis. Points above the diagonal indicate that SA produced a better solution than ACO on that instance; points below would indicate the opposite.

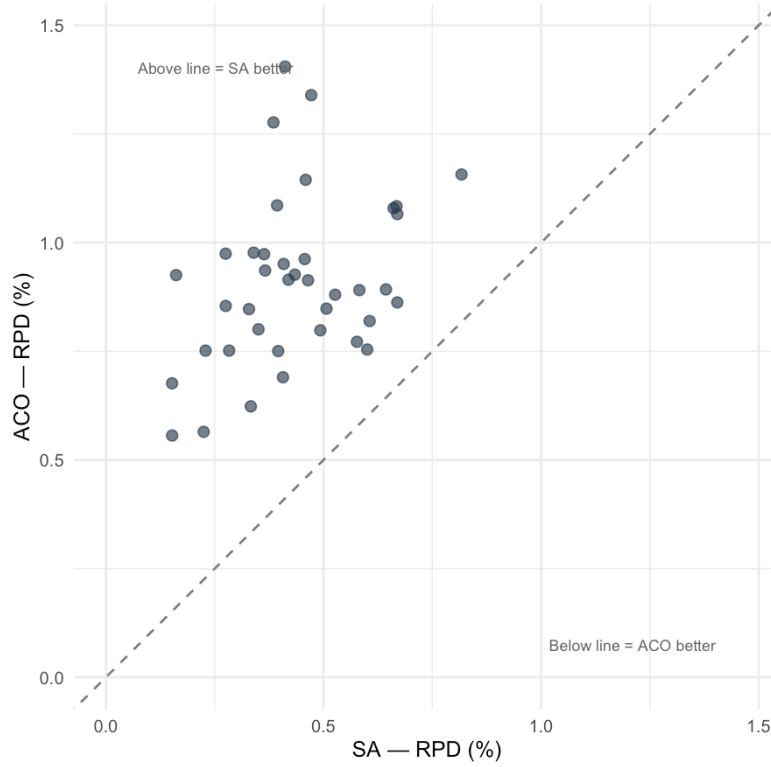


Figure 3: Correlation plot of RPD values: SA (x -axis) vs ACO (y -axis). Each point is one size-150 instance. The dashed line is $y = x$. Points below the line favour ACO; points above favour SA.

Figure 4 gives a per-instance view, sorted by the RPD obtained by SA.

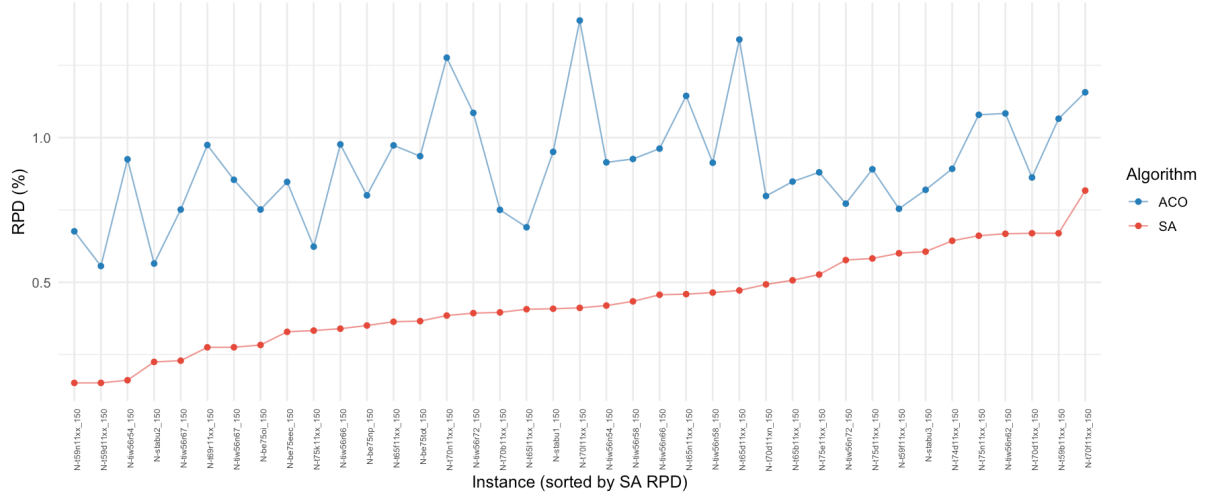


Figure 4: Per-instance RPD (%) for SA and ACO. Instances are ordered by increasing SA RPD.

4.1.2. Observations

The numerical results lead to three observations.

First, **Simulated Annealing is the strongest method**: it reaches an average RPD of 0.435% with a small standard deviation of 0.160 percentage points. ACO is roughly twice as far from the best-known solutions on average (0.910%) and is also slightly more dispersed.

Second, **both SLS methods clearly improve on plain VND**. The VND-TIE baseline from Exercise 1 stops at the first VND local optimum and reaches an average RPD of 1.690%, which is about 4 times worse than SA and about 2 times worse than ACO. This was expected: the SLS methods use 366.72 s per instance whereas VND finishes in less than half a second, but the experiment confirms that the extra time budget is effectively converted into solution quality.

Third, the **advantage of SA over ACO is highly consistent**: in Figure 3 every point lies above the diagonal, and in Figure 4 the SA curve is below the ACO curve on every single one of the 39 instances. There is no instance where ACO outperforms SA. This means the average RPD difference is not driven by a few outliers but is a systematic effect across the whole benchmark.

4.1.3. Statistical Tests

We apply the Wilcoxon signed-rank test at significance level $\alpha = 0.05$ to determine whether the differences observed above are statistically significant [4].

Algorithm A	Algorithm B	p-value	Significant?
SA	ACO (MMAS)	$< 10^{-7}$	Yes
SA	VND-TIE (Ex. 1)	$< 10^{-7}$	Yes
ACO (MMAS)	VND-TIE (Ex. 1)	$< 10^{-7}$	Yes

Table 3: Wilcoxon signed-rank test results ($\alpha = 0.05$) comparing the SLS algorithms and the best configuration from Exercise 1. Each test is paired over the 39 instances of size 150.

All three differences are significant at $\alpha = 0.05$ with p -values numerically indistinguishable from zero (the Wilcoxon statistic $W = 0$ for both SA-vs-ACO and SA-vs-VND, meaning that on

every paired instance the algorithm on the left was strictly better). We can therefore confidently rank the three approaches as $SA \prec ACO \prec VND-TIE$ with respect to RPD.

4.2. Exercise 2.1 — Run-Time Distributions

Run-time distributions (RTDs) measure how the probability of finding a solution of a given quality evolves over time. For each algorithm, we ran 25 independent repetitions on the first two instances of size 150 (N-be75eec_150 and N-be75np_150), with a cut-off time of $10 \times t_{\max} \approx 3667$ s. The target solution quality was set to within 0.5% of the best-known solution:

$$\text{target} = \text{BestKnown} \times (1 - 0.005)$$

Table 4 summarises the empirical success rates obtained at the cut-off, and Figure 5 shows the RTD curves side by side for the two instances.

Instance	Algorithm	Success rate (RPD $\leq 0.5\%$)	Mean RPD (%)
N-be75eec_150	SA	60%	0.431
N-be75eec_150	ACO	12%	0.571
N-be75np_150	SA	64%	0.458
N-be75np_150	ACO	12%	0.610

Table 4: Empirical success rates of SA and ACO over 25 independent runs on the first two size-150 instances. The target quality is $\text{RPD} \leq 0.5\%$.

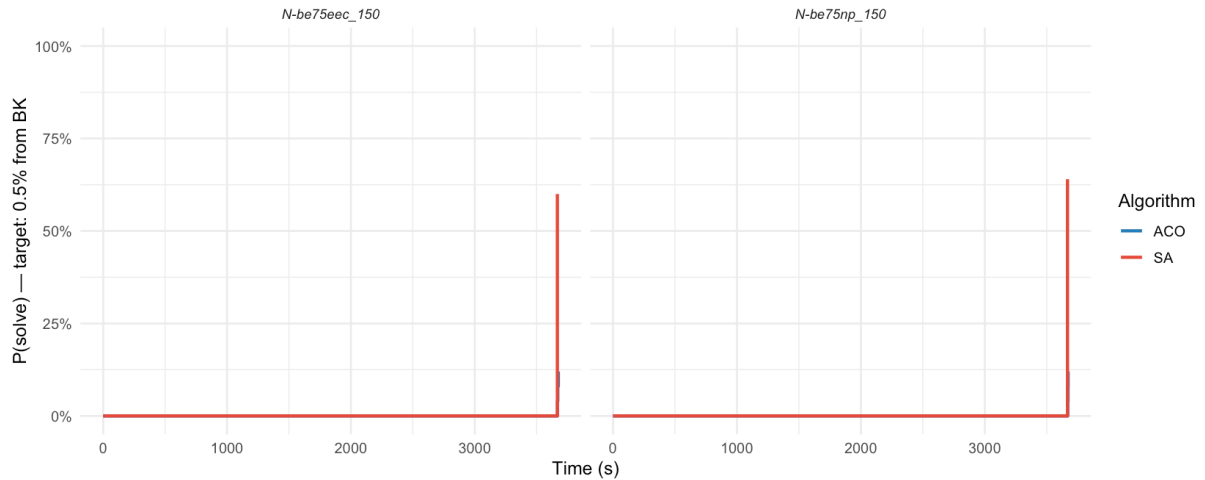


Figure 5: Run-time distributions on the two first size-150 instances. Each curve shows the fraction of the 25 runs whose final solution is within 0.5% of the best known, as a function of the elapsed time.

4.2.1. Observations

The two instances behave very similarly, which suggests that the trends below are robust rather than instance-specific.

The RTD curves are essentially flat at zero until the very end of the budget and then jump to their final value. This is a consequence of our implementation choice: both algorithms only report their final solution at the end of the run, so the RTD effectively reduces to the empirical **success rate at the cut-off**. With this caveat in mind, the comparison is still informative:

SA reaches the 0.5% target in roughly 60-64% of the runs, whereas ACO reaches it only in 12% of the runs on both instances. This is consistent with the per-instance results from Table 2 — SA’s average RPD is well below 0.5% while ACO’s is near 0.9%.

If the target is tightened to 0.25% from the best-known, SA still solves 20% of the runs on `N-be75eec_150` while ACO never reaches it; at 0.1% neither algorithm succeeds within the budget. This indicates that solving size-150 instances to near-optimality remains hard even with one hour of computation per run, and motivates the use of additional intensification or longer time budgets for higher-quality requirements.

5. Conclusion

We have implemented and evaluated two stochastic local search algorithms for the Linear Ordering Problem: a Simulated Annealing with insert moves and an automatically calibrated temperature schedule, and a MAX-MIN Ant System using VND-TEI as a per-ant local search. Both build on the VND machinery from the first implementation exercise and share the same time budget $t_{\max} \approx 367$ s per instance, calibrated so that each SLS run is roughly equivalent to 500 successive VND restarts.

On the 39 LOP instances of size 150, **Simulated Annealing was the clear winner** with an average RPD of 0.435% against 0.910% for ACO. Both methods strongly improve over plain VND-TIE from Exercise 1, which has an average RPD of 1.690%. The Wilcoxon signed-rank tests confirm that all pairwise differences are highly significant, and the per-instance comparison shows that SA dominates ACO on every single instance — there is no problem in the benchmark on which ACO is even tied with SA. The run-time distribution analysis on two representative instances reinforces this conclusion: at the 0.5% target, SA reaches the target in roughly 60% of the runs whereas ACO only reaches it in 12% of them.

Two reasons probably explain ACO’s weaker performance in this setup. First, each ant performs a full VND descent, which is expensive at $n = 150$ and limits the number of construction-update cycles that fit in the time budget; the colony only manages a handful of pheromone updates. Second, the construction probabilities use the row-sum heuristic with $\beta = 3$, which is already a strong attractor and may leave too little room for the pheromone to push the colony toward genuinely different regions. SA, on the other hand, performs millions of cheap random insert moves and benefits from the automatic temperature calibration, which removes the need for any per-instance tuning.

Overall, for this size of LOP instance and this time budget, **we would recommend Simulated Annealing**: it is simpler, has fewer parameters, calibrates itself, and consistently produces the best solutions of the three methods compared in this report.

References

- [1] S. Kirkpatrick, C. D. Gelatt, and M. P. Vecchi, *Optimization by Simulated Annealing*, vol. 220, no. 4598. 1983, pp. 671–680.
- [2] Wikipedia contributors, “Ant colony optimization algorithms.” [Online]. Available: https://en.wikipedia.org/wiki/Ant_colony_optimization_algorithms
- [3] T. Stützle and H. H. Hoos, “MAX-MIN Ant System,” *Future Generation Computer Systems*, vol. 16, no. 8, pp. 889–914, 2000.

- [4] Wikipedia contributors, “Wilcoxon signed-rank test.” [Online]. Available: https://en.wikipedia.org/wiki/Wilcoxon_signed-rank_test